

Revisiting Vital Signs Inference Across Subjects with COTS mmWave Radar

Areesh Sobhani
Purdue University
West Lafayette, IN, USA
asobhani@purdue.edu

Qiuye He
University of Missouri–Kansas City
Kansas City, MO, USA
qiuye.he@umkc.edu

Yugyung Lee
University of Missouri–Kansas City
Kansas City, MO, USA
leeyu@umkc.edu

Abstract—Vital signs such as respiration and heart rate are fundamental indicators of health, supporting early diagnosis, continuous patient care, and the detection of disorders such as sleep apnea, asthma, and cardiovascular disease. Traditional monitoring methods, including airflow sensors, chest straps, and photoplethysmography (PPG) devices, require direct contact with the body and are often uncomfortable or impractical for long-term use. Contactless sensing provides an attractive alternative, and among various modalities—optical, audio, Wi-Fi RSSI/CSI—millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar has emerged as particularly promising due to its high resolution and sensitivity to subtle chest-wall micromotions. In this work, we revisit vital-sign inference across subjects using commercial off-the-shelf (COTS) 77 GHz mmWave radar. We design a signal processing pipeline that decomposes intermediate-frequency (IF) returns into range, Doppler, and phase domains to isolate respiration- and heartbeat-induced micromotions. Through phase unwrapping, filtering, and statistical estimation, we derive both respiration rate and heart rate without requiring subject-specific calibration. In a pilot study with diverse participants and postures, the proposed system achieved stable respiration-rate estimation and heart-rate inference with mean absolute errors of 0.95 breaths/minute and 2.00 beats/minute, respectively. These results demonstrate the feasibility of robust, cross-subject physiological monitoring with commodity mmWave radar, highlighting its potential for scalable, low-cost, and unobtrusive healthcare applications.

Index Terms—Millimeter wave (mmWave) radar, Frequency modulated continuous wave (FMCW), biometric inference, vital sign monitoring, contactless sensing

I. INTRODUCTION

Monitoring vital signs is routine practice in healthcare, where changes in parameters such as heart rate and respiration rate often serve as early indicators of serious medical conditions. Early detection of abnormalities in these signals has the potential to improve survival rates for conditions including sleep apnea, asthma, pneumonia, cardiovascular disease, and chronic obstructive pulmonary disease [1]–[5]. While vital-sign monitoring is critical to clinical practice, there is also growing public interest in self-monitoring, driven by the proliferation of wearable devices specialized in heart rate tracking [6]–[9]. Although wearable and attachable devices are widely adopted, they suffer from several important limitations. Specifically, long-term use may cause discomfort or even skin irritation, they require continuous contact with the body to function properly, and in some cases they may interfere with natural physiological patterns.

Contactless and unobtrusive monitoring offers a compelling alternative. Unlike wearable devices, wireless sensing technologies can track multiple vital signs simultaneously without user burden, and they can be applied in diverse contexts ranging from home environments to hospitals and nursing facilities [10]–[26]. For example, radar-based monitoring eliminates the tactile discomfort commonly associated with wearable devices as well as the detachment risk of wired electrodes, making it particularly appealing for applications such as sleep monitoring, chronic disease management, and critical care [17], [19], [27]–[34]. By sensing the periodic micromotions of the chest wall and skin surface induced by respiration and cardiovascular pulsations, radar systems can noninvasively estimate both respiration rate and heart rate with high temporal resolution.

A variety of non-contact approaches have been explored. Optical methods extract respiration from video streams [35]–[37], but they are sensitive to lighting and line-of-sight conditions. Audio-based approaches use microphones to capture breathing sounds [38], [39], though they degrade significantly in noisy environments or at longer distances. Wi-Fi-based schemes leveraging received signal strength indicator (RSSI) or channel state information (CSI) have also been investigated, offering the advantage of reusing ubiquitous communication hardware. RSSI-based methods can detect coarse respiratory patterns, but their accuracy is limited by multipath fading and channel noise [23], [40]–[42]. CSI-based methods, by contrast, provide fine-grained magnitude and phase measurements that enable more robust detection of breathing-induced micromotions [21], [22], [25], [43]–[49]. While promising, existing CSI approaches are often constrained by specific experimental setups, require large training datasets, or lack generalization across subjects.

In contrast, radio-frequency (RF) radar technologies such as Doppler radar [27]–[30], ultra-wideband (UWB) radar [19], [31], [32], and frequency-modulated continuous-wave (FMCW) radar [17], [33], [34] have demonstrated high reliability in capturing both respiratory and cardiac signals. Doppler radar exploits phase shifts caused by the periodic displacement of the chest wall, offering a relatively simple hardware implementation; however, it is sensitive to null-point effects and interference from unwanted motion. UWB radar, by contrast, transmits short pulses over a very wide spectrum

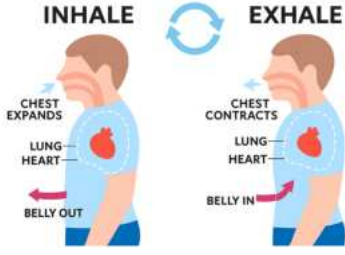


Fig. 1: Chest-wall vibrations are periodic motions caused by both breathing and heartbeat.

and reconstructs channel impulse responses (CIRs) with fine temporal resolution, enabling detection of subtle physiological micromotions even in multipath-rich environments. FMCW radar extends these capabilities by simultaneously providing range and velocity discrimination. FMCW radar achieves fine range resolution and high phase sensitivity, thereby supporting robust extraction of respiration and heart rate across diverse distances, orientations, and postures. More recent deployments in assisted-living facilities further confirm the feasibility of low-cost millimeter-wave (mmWave) radar (e.g., operating at a frequency about 77 GHz) modules for improving accessibility and patient comfort in real-world healthcare settings [50]–[54].

Building on these advances, this work revisits the problem of mmWave-based vital-sign inference and investigates the feasibility of cross-subject monitoring using commercial off-the-shelf (COTS) radar hardware. We design a signal processing pipeline that integrates conventional radar-domain transformations—range FFT, range-Doppler analysis, and phase tracking—with advanced filtering and statistical estimation techniques to robustly isolate subtle physiological micromotions. In contrast to prior approaches that require subject-specific calibration, our framework emphasizes generalization across individuals, which is a critical step toward scalable and widely deployable systems. Through pilot experiments with diverse participants and realistic office-like conditions, we demonstrate that COTS mmWave radar achieves stable and accurate estimation of both respiration rate (RR) and heart rate (HR). Specifically, our experimental results show that the system estimates respiration rate with a mean absolute error below 0.95 breaths per minute (bpm) and heart rate with an error below 2.00 beats per minute (bpm) for seated subjects. These findings reinforce the potential of mmWave radar as a cost-effective, unobtrusive, and practical solution for continuous vital-sign monitoring, while highlighting its suitability for real-world healthcare and assisted-living applications. The main contributions of this work are summarized as follows:

- We revisit the problem of vital-sign inference and present a framework for cross-subject respiration and heart rate monitoring using COTS mmWave radar.
- We design a signal processing pipeline that combines range FFT, range-Doppler analysis, phase tracking, and advanced filtering for robust extraction of physiological micromotions.

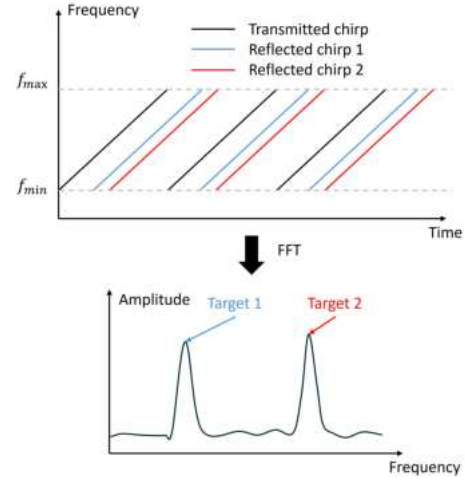


Fig. 2: The peaks correspond to targets detected after applying the range FFT.

- We conduct real-world experiments with diverse participants and postures, demonstrating accurate cross-subject estimation without subject-specific calibration.
- We provide a quantitative evaluation, showing stable respiration-rate and heart-rate estimation with mean absolute errors below 0.95 breaths per minute and 2.00 beats per minute, respectively.

The rest of this paper is organized as follows. Section II introduces the preliminaries of mmWave radar sensing. Section III presents the methodology for vital-sign inference. Section IV reports the experimental evaluation and analysis. Section V reviews related work on wireless vital-sign inference techniques. Finally, Section VI concludes the paper.

II. BACKGROUND

A. Chest Vibrations

Figure 1 shows that the chest-wall micromotions are generated by respiration and by the heartbeat. Because breathing and cardiac activity are periodic, the resulting chest-wall vibrations are likewise periodic. A mmWave radar measures these vibrations by transmitting a high-frequency signal and receiving its echo from the chest. Tiny chest displacements change the round-trip path length, imprinting periodic phase modulation on the radar’s intermediate-frequency (IF) signal. Even when chest displacement may not be visually discernible, the radar resolves these phase modulations and therefore infers the underlying respiration and heartbeat. In the next section, we detail the mmWave hardware that enables this capability.

B. FMCW Fundamentals

Frequency-Modulated Continuous Wave (FMCW) transmits a sinusoid whose instantaneous frequency increases linearly in time (a *chirp*). A baseband model of one up-chirp is

$$s_{tx}(t) = \exp\{j2\pi(f_0t + \frac{1}{2}\mu t^2)\}, \quad t \in [0, T_c], \quad (1)$$

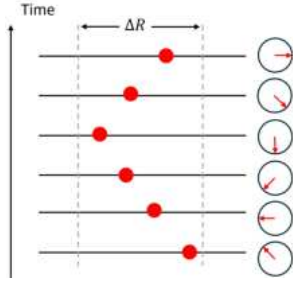


Fig. 3: The phase of the range-FFT peak tracks the object's oscillatory motion.

where f_0 is the start frequency, μ the chirp slope, and T_c the chirp duration. For a point target at range R and radial velocity v , the round-trip delay is $\tau = 2R/c$, and the two-way Doppler shift is $f_D = 2v/\lambda$ with wavelength $\lambda = c/f_c$.

After dechirping (mixing s_{rx} with s_{tx} and low-pass filtering), the intermediate-frequency (IF) signal is well approximated by

$$s_{IF}(t) \approx A \exp\{j2\pi(f_b t + \phi_0)\}, \quad (2)$$

where the beat frequency on an up-chirp is $f_b \approx \mu\tau + f_D$. Using up/down-chirp pairs can help separate range and Doppler contributions. Range is estimated from the fast-time beat as

$$R \approx \frac{c}{2\mu} f_b, \quad (3)$$

for stationary targets, while velocity is inferred from Doppler-induced phase or frequency shifts across successive chirps (slow time).

As illustrated in Figure 2, the radar receiver collects multiple reflected chirps arising from different multipath components in the environment. Applying a fast-time FFT to $s_{IF}(t)$ resolves these multipath components and reveals their corresponding ranges. The achievable range resolution is determined by the chirp bandwidth $B = \mu T_c$, given by

$$\Delta R = \frac{c}{2B}, \quad (4)$$

while the Doppler resolution is determined by the length of the coherent slow-time aperture across successive chirps.

However, even when the maximum operational bandwidth of the radar (approximately 4 GHz) is utilized by a chirp, the achievable range resolution is about 3.75 cm. This resolution is insufficient for detecting micrometer-level vibrations, such as those induced by chest-wall motion.

To accurately track micromotion that remains within a single range bin (i.e., displacements smaller than the range resolution), we exploit the phase of the reflected signal at the target's range bin. As shown in Figure 3, phase variations provide a highly sensitive measure of minute displacements of the reflector, expressed as

$$\Delta\phi = \frac{4\pi\Delta r}{\lambda}, \quad (5)$$

where Δr is the one-way displacement of the reflector and λ is the wavelength of the transmitted signal. With $\lambda \approx 4$ mm and

a typical phase noise of 0.057° , the minimum resolvable displacement is approximately $\Delta r_{\min} \approx 0.32 \mu\text{m}$ by continuously tracking phase variations.

C. System Parameters

Expanding upon the high-level overview in Section II-B, this section provides a detailed description of the radar system parameters and their role in vital-sign extraction.

At the chirp level, each transmitted waveform defines the fundamental sensing resolution and phase sensitivity of the system. The relevant parameters are:

- **Start Frequency (f_0):** The initial frequency of the transmitted chirp, here set to 77 GHz. Operating at mmWave frequencies reduces the wavelength ($\lambda = 3.9$ mm), which enhances phase sensitivity to sub-millimeter chest-wall displacements caused by respiration and heartbeat.
- **Chirp Duration (T_c):** The active time of one chirp ramp, configured as 50 μs . Longer chirp durations increase frequency resolution but reduce the temporal resolution of phase sampling.
- **Chirp Slope (μ):** The linear frequency sweep rate, set to 70 MHz/ μs . A steeper slope increases the bandwidth for a given chirp duration, thereby improving range resolution and the ability to separate the subject from background clutter.
- **Bandwidth ($B = \mu T_c$):** With the chosen slope and chirp duration, the effective bandwidth is 3.5 GHz. According to radar theory, range resolution is inversely proportional to bandwidth, and this wide bandwidth supports accurate localization of the subject in near-field monitoring.
- **Center Frequency (f_c):** The mean frequency of the chirp (78.75 GHz), which determines the effective wavelength (3.8 mm). The shorter wavelength increases phase sensitivity, enabling the detection of heartbeat-level micromotions.
- **ADC Sampling Rate (f_s):** The rate at which intermediate-frequency (IF) signals are digitized, set to 4 MHz. Higher sampling rates allow faithful reconstruction of wideband chirps but also increase data throughput.

At the frame level, groups of chirps define the temporal window over which signals are aggregated and processed. The corresponding parameters are:

- **Chirps per Frame (N_c):** Each frame consists of 1024 chirps. Larger N_c values improve frequency estimation accuracy but increase frame duration and latency.
- **Number of Frames:** We capture 400 frames per measurement session. This defines the overall observation period and thus the stability of the averaged estimates of respiration and heart rate.
- **Frame Duration:** Each frame lasts 0.05 s, set by the chirp cycle time and the number of chirps per frame. This duration dictates how frequently respiration and heartbeat features can be updated.

Together, these chirp- and frame-level parameters yield a system configuration that provides millimeter-level sensitivity

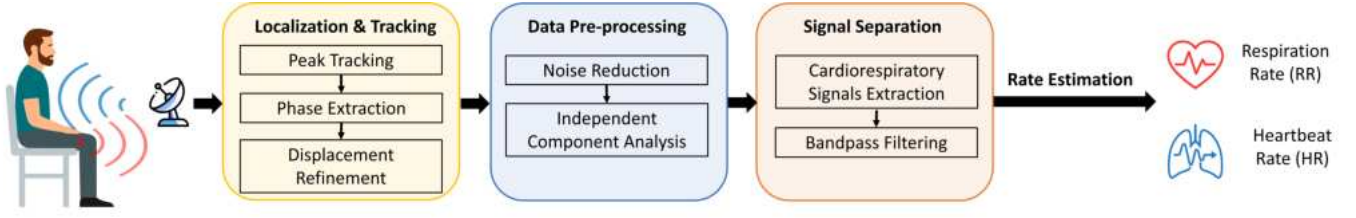


Fig. 4: System overview of the proposed scheme.

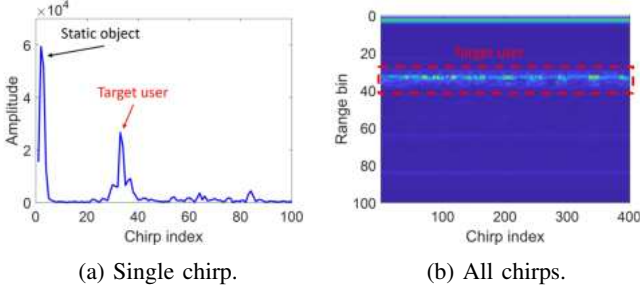


Fig. 5: Range-FFT results.

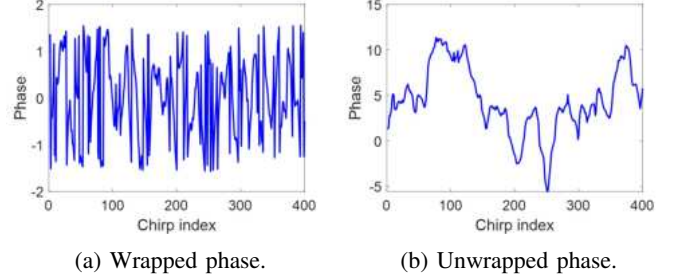


Fig. 6: Phase unwrapping.

to chest motion and sufficient temporal resolution to track both respiration and heartbeat rates.

III. METHODOLOGY

A. System Overview

The system overview is illustrated in Figure 4. The mmWave transceiver emits FMCW chirps toward the target subject and receives the reflected signals. After demodulation, the signals are processed by the *Localization & Tracking* module, which comprises three components: peak localization, phase tracking, and displacement refinement. Peak localization identifies the range-FFT peaks corresponding to the target, phase tracking monitors the temporal phase variations of these peaks, and displacement refinement enhances the displacement estimates induced by chest-wall micromotions. Subsequently, the mmWave responses are fed into a *Data Pre-processing* module, which reduces noise first and then removes uncorrelated components using Independent Component Analysis (ICA). The processed data then enter the *Signal Separation* module, where the cardiorespiratory signals comprising both respiration and heartbeat components are extracted. These components are separated using band-pass filters designed for their respective frequency ranges. Finally, the respiration rate (RR) and heart rate (HR) are estimated through frequency-domain analysis of the separated signals.

B. Localization & Tracking

1) *Peak Localization*: As discussed in Section II-B, a range-FFT window includes not only reflections from the human chest but also contributions from static objects (e.g., walls, furniture) and multipath components. In the single-chirp example shown in Figure 5a, two dominant peaks are observed. To identify the peaks corresponding to the human

chest, our system tracks peaks consistently across successive frames, as illustrated in Figure 5b. Noisy peaks fluctuate randomly in range and amplitude and are therefore naturally discarded during temporal tracking. Similarly, peaks originating from static objects remain stable but do not exhibit the periodic micromotions associated with respiration and heartbeat. By enforcing both temporal consistency and periodic modulation, the system robustly isolates the chest reflection for subsequent processing.

2) *Phase Extraction*: Since the range resolution of our system (approximately 3.75 cm for a 4 GHz bandwidth) is insufficient to directly resolve the micro-vibrations of the chest wall, we instead leverage the phase variations of the radar returns. Figure 6 shows the phase signal obtained from the selected range-bin peak. Due to the short wavelength of mmWave signals, sub-millimeter chest displacements induce rapid phase changes. As a result, the measured phase is subject to wrapping, where values exceeding the interval $[-\pi, \pi]$ are folded back into this range, creating artificial discontinuities, as illustrated in Figure 6a. To recover the true motion, we apply phase unwrapping, which detects these discontinuities and compensates for them by adding integer multiples of 2π , thereby reconstructing a continuous phase trajectory. The resulting unwrapped phase, corresponding to the chest reflection bin, is presented in Figure 6b.

3) *Displacement Refinement*: Accordingly, the chest-wall displacement d can be estimated from the phase variation of the reflected signal. Specifically, the relationship is given by

$$d = \frac{\lambda}{2\pi} \phi,$$

where ϕ represents the phase change of the received signal and λ denotes the wavelength of the mmWave carrier.

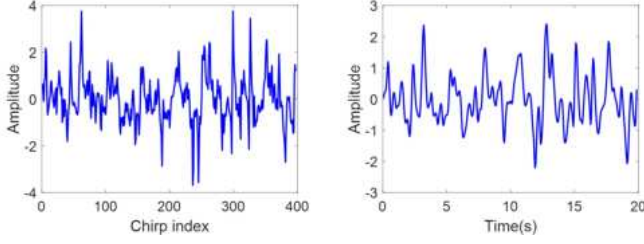


Fig. 7: Signals after Fast independent component analysis.

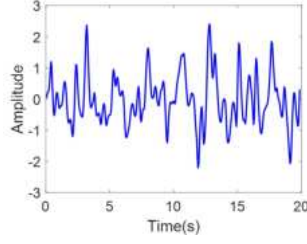
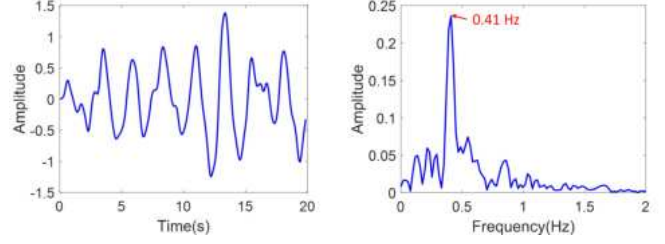


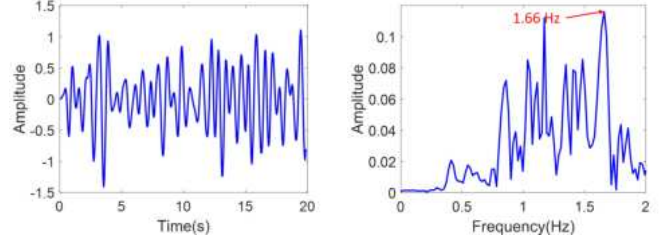
Fig. 8: Cardiorespiratory signals extraction.



(a) In the time domain.

(b) In the frequency domain.

Fig. 9: Respiration signal.



(a) In the time domain.

(b) In the frequency domain.

Fig. 10: Heartbeat signal.

C. Data Pre-processing

1) *Noise Reduction*: The extracted chest-displacement signal may contain spurious spikes arising from hardware artifacts or abrupt interference, which typically manifest as impulse noise. To address this issue, we design a dedicated filter for impulse-noise suppression. The core idea is to detect samples whose local differences with their neighbors exceed a predefined threshold in both the forward and backward directions. For each current sample, the filter evaluates its difference with the preceding and following samples; if both differences exceed the threshold with the same polarity, the sample is classified as an impulse outlier. In such cases, its value is replaced by a linear interpolation between its two neighbors to preserve continuity. Otherwise, the original sample is retained.

2) *Independent Component Analysis (ICA)*: To further separate the cardiorespiratory signals from clutter and noise, we employ Fast Independent Component Analysis (Fast ICA) [55]. Fast ICA is a blind source separation algorithm that decomposes multivariate observations into additive, statistically independent components. The received radar signals consist of a mixture of reflections from the chest, surrounding static objects, and environmental noise. Since respiration and heartbeat induce quasi-periodic micromotions that are statistically independent of static clutter and random noise, Fast ICA provides an effective way to identify the physiological signals of interest. Figure 7 illustrates the separated signals obtained after applying Fast ICA.

D. Signal Separation

1) *Cardiorespiratory Signals Extraction*: The preprocessed radar signals contain not only chest-induced micromotions but also a variety of undesired components, such as low-frequency baseline drift and high-frequency noise. To extract the physiological activity of interest, we apply a band-pass filter that retains only the frequency range associated with cardiorespiratory motion, namely 0.1–2 Hz. This filtering step attenuates out-of-band interference and static artifacts while preserving the quasi-periodic vibrations generated by chest-wall displacement. As a result, the filtered output provides a clearer representation of the underlying cardiorespiratory activity, as illustrated in Figure 8, and serves as a robust input for subsequent vital-sign analysis.

2) *Bandpass Filtering*: In particular, respiration typically appears in the lower frequency range of 0.1–0.6 Hz (corresponding to 6–36 breaths per minute), while the heartbeat signal lies in the higher frequency range of 0.8–2.0 Hz (corresponding to 48–120 beats per minute). Band-pass filters designed for these ranges effectively suppress out-of-band interference while retaining the quasi-periodic micromotions induced by respiration and heartbeat. Figure 9 illustrates the filtered respiration signal in both the time and frequency domains, with a dominant frequency component of 0.41 Hz. Similarly, Figure 10 shows the filtered heartbeat signal in both domains, with a dominant frequency component of 1.66 Hz. This filtering stage enables a clear separation of the two vital-sign signals and provides a reliable basis for accurate estimation of both respiration rate (RR) and heart rate (HR).

E. Rate Estimation

After identifying the dominant frequency components of the respiration and heartbeat signals, the corresponding physiological rates are obtained by converting frequency (Hz) into rate (cycles per minute). Specifically, the respiration rate (RR) is calculated as

$$RR = f_{\text{resp}} \times 60,$$

where f_{resp} is the dominant respiratory frequency in hertz. Likewise, the heart rate (HR) is estimated as

$$HR = f_{\text{heart}} \times 60,$$

where f_{heart} denotes the dominant cardiac frequency in hertz. For example, a detected respiratory frequency of 0.41 Hz corresponds to approximately 25 breaths per minute, while a heartbeat frequency of 1.66 Hz corresponds to 100 beats per

TABLE I: Radar system parameters used in experiments.

Parameter	Value
Propagation speed c	3×10^8 m/s
Start frequency f_0	77 GHz
Chirp duration T_c	50 μ s
Chirp slope μ	70 MHz/ μ s
Bandwidth $B = \mu T_c$	3.5 GHz
Center frequency f_c	78.75 GHz
Wavelength $\lambda = c/f_c$	3.8 mm
ADC sampling rate f_s	4 MHz
Chirps per frame N_c	1024
Number of frames	400
Minimum frame duration	0.05 s
Target distance	0.305 m (1 ft)

minute. This frequency-to-rate conversion provides a direct and reliable method for quantifying vital signs from the mmWave radar signals.

IV. EXPERIMENTAL EVALUATION

We implement our vital-sign inference framework on the Texas Instruments (TI) IWR1843BOOST mmWave radar platform [56], which integrates a 77 GHz FMCW transceiver with on-chip digital signal processing (DSP) units and hardware accelerators. For high-speed data acquisition, the radar is paired with the TI DCA1000EVM capture card [57], which streams raw intermediate-frequency (IF) samples from the radar front-end to a host computer for offline processing. This configuration enables multi-channel radar data to be recorded in real time while avoiding memory and computational constraints of the on-board processor. The key radar system parameters employed in both simulation and physical experiments are summarized in Table I, providing a reproducible baseline for future studies.

A. Evaluation Setup

To evaluate the accuracy and robustness of the proposed mmWave-based vital-sign inference system, we conducted experiments with eight volunteers of varying sex, height, weight, and body build over one month. Each subject also wore a Masimo MightySat Fingertip Pulse Oximeter [7], a hospital-grade device used to obtain ground-truth measurements of breathing rate and heart rate. All participants provided informed consent prior to data collection. The demographic characteristics of the subjects are summarized in Table II.

We evaluated the system in a typical indoor scenario, specifically an office room where each subject was seated in a chair, as shown in Figure 11. Each subject performed 5 trials under different activity levels, including rest, mild exertion, and deep-breathing cycles. To quantify performance, we employed the following two metrics:

- **Absolute Error (AE):** defined as the absolute difference between the estimated rate and the ground truth,

$$AE = |\hat{r} - r_{\text{true}}|,$$

TABLE II: Demographic profiles of subjects used in experiments.

User	Sex	Height	Weight (lbs)
U1	Female	5'0"	146
U2	Female	5'4"	131
U3	Male	5'7"	154
U4	Male	6'1"	204
U5	Female	5'5"	127
U6	Male	5'8"	176
U7	Male	5'10"	185
U8	Female	5'2"	132

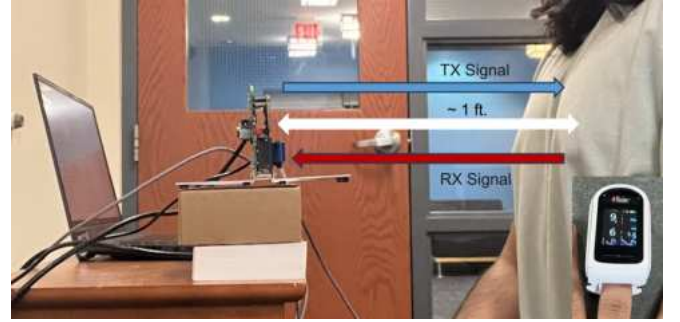


Fig. 11: Evaluation setup.

where $\hat{r}$ is the estimated respiration or heart rate, and r_{true} is the ground-truth value.

- **Percentage Error (PE):** computed as

$$PE (\%) = \frac{|\hat{r} - r_{\text{true}}|}{r_{\text{true}}} \times 100,$$

where $\hat{r}$ is the estimated rate, and r_{true} is the corresponding ground-truth measurement.

B. Overall Performance

1) *Respiratory Rate Inference:* The statistical results are summarized in the boxplots shown in Figure 12 and Figure 13, which illustrate the distributions of AE and PE for respiratory rate inference. The mean absolute error (AE) was approximately 1-2 bpm, with the median error closely aligned to the mean, indicating stable performance without large outliers. The percentage error (PE) for respiratory rate remained below 10-15% in most cases, demonstrating robustness across varying conditions. These results confirm that the system provides consistent performance across subjects and trials, with minimal extreme deviations.

2) *Heart Rate Inference:* The statistical results are summarized in the boxplots shown in Figure 12 and Figure 13, which illustrate the distributions of AE and PE for heart rate inference. The mean AE was slightly higher than that of RR but remained within 3-4 bpm, with a relatively low standard deviation, confirming the reliability of the estimates. The PE for HR was generally below 8-12%, even in higher-rate scenarios, highlighting the ability of the system to maintain accuracy across a broad physiological range.

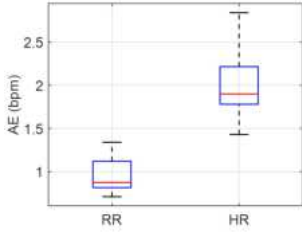


Fig. 12: AE for both RR and HR inference.

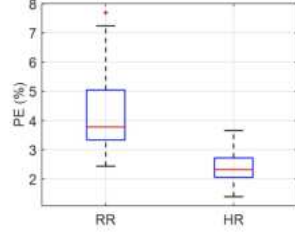


Fig. 13: PE for both RR and HR inference.

TABLE III: Statistical results of RR estimation (Absolute Error (AE) in bpm, Percentage Error (PE) in %).

User	AE _{min}	AE _{max}	AE _{mean}	PE _{min}	PE _{max}	PE _{mean}
U1	0.77	1.17	0.96	5.68	7.70	6.38
U2	0.87	1.34	1.03	3.85	7.24	5.03
U3	0.76	1.05	0.87	2.70	4.40	3.40
U4	0.75	1.07	0.88	2.65	3.56	3.13
U5	0.72	1.17	0.87	3.70	6.52	4.73
U6	0.78	1.14	0.94	2.44	4.15	3.19
U7	0.75	1.28	1.04	2.57	4.12	3.26
U8	0.71	2.18	1.23	3.67	11.66	6.43

Overall, these findings confirm that the proposed system can robustly infer both respiration and heart rates with high accuracy, validating its suitability for non-contact vital-sign monitoring applications.

C. User Study

We recruited eight volunteers and asked each participant to breathe normally in an office environment. Each subject performed five repeated trials for data collection.

Table III and Table IV summarize the statistical performance of the proposed system in estimating respiration rate (RR) and heart rate (HR), respectively, across eight users. We use $Q_{\min}$, $Q_{\max}$, and Q_{mean} , where $Q \in \{\text{RR}, \text{HR}\}$, to denote the minimum, maximum, and mean values of the respiration rate (RR) and heart rate (HR), respectively. For RR estimation, the absolute error (AE) remained consistently low, with mean values below 1.3 bpm for all subjects. The lowest mean AE was observed for User 3 (0.87 bpm), while the highest occurred for User 8 (1.23 bpm), which also showed the largest maximum AE of 2.18 bpm. The percentage error (PE) values for RR estimation were generally within 3–7% for most users, with User 8 again showing a higher variability ($\text{PE}_{\max} = 11.66\%$). These results confirm that the system is capable of tracking respiration with high accuracy, with errors well within clinically acceptable ranges.

For HR estimation (Table IV), the mean AE values were slightly higher compared to RR, ranging from 1.82 to 2.26 bpm across users. Importantly, the maximum AE remained below 3 bpm for all subjects, demonstrating stable estimation performance even in higher frequency ranges. The mean PE values ranged between 1.92% and 3.19%, with minimal variability across users, highlighting the robustness of the system for HR detection. The relatively narrow spread

TABLE IV: Statistical results of HR estimation (Absolute Error (AE) in bpm, Percentage Error (PE) in %).

User	AE _{min}	AE _{max}	AE _{mean}	PE _{min}	PE _{max}	PE _{mean}
U1	2.16	2.37	2.26	2.91	3.66	3.19
U2	1.77	2.32	2.00	2.32	2.90	2.61
U3	1.73	2.55	1.93	1.98	2.89	2.25
U4	1.47	2.23	1.91	1.58	2.45	2.14
U5	1.78	1.89	1.82	2.08	2.53	2.27
U6	1.78	2.84	2.07	1.85	2.98	2.19
U7	1.56	2.45	1.98	1.39	2.21	1.92
U8	1.43	2.62	2.04	1.86	3.08	2.48

of AE and PE in the boxplots (not shown here) further supports the consistency of HR estimation.

Overall, the analysis of both RR and HR results indicates that the proposed mmWave-based vital sign inference system can reliably estimate cardiorespiratory activity across diverse subjects, achieving mean errors well within 2 bpm for RR and 3 bpm for HR. These findings validate the suitability of the approach for non-contact, continuous monitoring applications in realistic indoor environments.

D. Impact of Gender

Despite these differences in baseline vital signs, the proposed mmWave-based estimation system demonstrated comparable accuracy across genders. For respiration rate (RR), female subjects achieved slightly lower mean absolute error (AE) values, while the percentage error (PE) distributions for both genders largely overlapped, indicating similar robustness. For heart rate (HR), both female and male groups achieved mean AE values around 2 bpm and mean PE values below 3%, with no systematic bias observed between genders. These findings confirm that the system can reliably estimate vital signs across diverse physiological baselines, underscoring its generalizability for real-world deployment.

V. DISCUSSIONS

Although the proposed mmWave-based vital-sign monitoring framework demonstrates promising results, several limitations should be acknowledged, which also highlight opportunities for future research.

Diverse scenarios: The evaluation was conducted with a relatively small number of participants under controlled office-like conditions. Such a limited sample size and constrained environment do not fully reflect the diversity of real-world deployments, where user demographics, body morphologies, and environmental layouts may vary substantially. Future studies will expand experiments to include larger and more diverse cohorts, thereby enabling more rigorous statistical validation across age, gender, and body types.

Complex activities: The experimental protocol primarily focused on static, seated postures. The system's performance during dynamic activities such as walking, speaking, or sleeping remains unexplored. Extending evaluation to naturalistic scenarios such as free movement during daily activities, long-term sleep monitoring, and speech-related chest motion will be critical to assess robustness under realistic conditions.

Multi-user monitoring: The proposed system has so far been validated only in single-user scenarios. In practical settings, multiple individuals may share the same environment, leading to overlapping radar returns and inter-user interference. Future work will investigate multi-user monitoring techniques, including spatial filtering, advanced beamforming, and source separation methods, to reliably distinguish and track multiple subjects simultaneously.

Through-wall monitoring: The current framework assumes line-of-sight conditions. In through-wall or non-line-of-sight (NLoS) scenarios, signal attenuation and multipath propagation may significantly degrade performance. Investigating the feasibility of through-wall monitoring, for example through high-gain antenna design or enhanced signal processing, will be an important extension for applications in elder care, rescue, and security.

Computational efficiency: The current signal calibration and preprocessing pipeline is computationally intensive, which may limit real-time operation on embedded or resource-constrained devices. Future research will focus on optimizing algorithmic complexity and implementing efficient hardware acceleration to enable deployment on edge devices, thereby supporting real-time, low-power, and scalable operation in practical healthcare and smart-home settings.

Advanced algorithms: The integration of advanced signal processing with machine learning and deep learning models offers a promising avenue for improving robustness and accuracy in the presence of multipath interference, clutter, and dynamic activity. Data-driven approaches may further enable adaptive, environment-aware monitoring that generalizes across a wide variety of use cases.

Overall, while several challenges remain, this work reinforces the potential of mmWave radar as a practical platform for continuous, unobtrusive, and scalable vital-sign monitoring in real-world settings.

VI. RELATED WORK

Generally, existing wireless breathing rate inference techniques fall into the following categories:

RSS-based: Variations in received signal strength (RSS) across wireless links have been explored for breathing rate estimation [23], [40]–[42]. Breathing-induced chest movements slightly alter the propagation channel, leading to detectable fluctuations in RSS measurements. Several studies have demonstrated that RSS-based methods can achieve reasonable accuracy in short-range scenarios. For example, [42] places a mobile device directly on the chest to collect RSS signals and infer breathing patterns. Such approaches are simple and require no specialized hardware beyond commodity wireless devices. However, RSS provides only coarse-grained channel information and is strongly affected by noise and environmental dynamics. Moreover, these methods are only reliable when the target remains close to the receiver, which limits their applicability in many practical settings. In adversarial scenarios such as passive eavesdropping, where the

observer may prefer to remain far from the target to avoid detection, RSS-based monitoring is not feasible.

CSI-based: Channel state information (CSI) offers fine-grained channel measurements by capturing the amplitude and phase response at the subcarrier level in OFDM-based systems. Unlike RSS, which aggregates power across the entire channel, CSI retains frequency-selective details that are more sensitive to subtle physiological movements. As a result, CSI-based methods are capable of detecting breathing activity even at a distance or in non-line-of-sight conditions. This makes CSI particularly attractive for unobtrusive respiration monitoring in smart homes and healthcare environments [21], [22], [25], [43]–[49]. Recent empirical evidence [19] has shown that CSI-based methods provide more robust respiration estimates compared to UWB radar or RSS-based systems, due to their ability to leverage multipath diversity and capture finer temporal variations. Nonetheless, CSI acquisition often requires access to low-level channel data that is not always available on commodity devices, and signal calibration remains a challenge. Despite these limitations, CSI has emerged as one of the most promising techniques for contactless breathing monitoring.

Ultra-wideband (UWB) radar based: UWB radar transmits extremely short pulses over a very wide frequency spectrum, which enables fine time resolution and accurate ranging capabilities. The expansion and contraction of the chest cavity during respiration induce subtle changes in the multipath profile of transmitted signals. These changes can be captured by analyzing UWB channel impulse responses (CIRs) to infer breathing rate [19], [31], [32]. Due to its high temporal resolution, UWB radar can differentiate between multiple reflectors and is relatively robust in cluttered environments. However, UWB transmissions require a large allocated frequency bandwidth [58], which can limit deployment in practice due to regulatory constraints. In addition, UWB receivers must sample at very high rates and perform complex signal processing for CIR reconstruction, leading to high implementation cost and system complexity [59].

Doppler radar based: Doppler radar has been extensively studied for non-contact respiration monitoring [27]–[30]. According to Doppler theory, when a target undergoes periodic movement with zero average velocity, the reflected signal is phase modulated in proportion to the displacement [60]. The chest and abdominal regions of a stationary person act as such targets, with their periodic expansion and contraction generating micro-Doppler signatures. These phase variations can be processed to extract respiration patterns. Doppler radars are attractive because they typically require simpler transceiver hardware compared to UWB and can operate in continuous-wave mode with relatively low bandwidth requirements. However, Doppler-based methods suffer from the well-known “null point” problem, where certain target positions or orientations cause destructive interference in the received signal. This results in blind spots and reduced measurement accuracy [29], [61], making them less robust in realistic environments.

Frequency-Modulated Continuous Wave (FMCW) radar based: FMCW radar simultaneously provides range and veloc-

ity information by transmitting chirp signals whose frequency increases linearly with time. For respiration monitoring, the periodic displacement of the chest alters the propagation path length, which in turn modifies the reflection time and introduces measurable phase variations. By analyzing these variations, the breathing rate can be inferred [17], [33]. Compared to Doppler radar, FMCW offers additional range discrimination, which allows the system to isolate reflections from the human body even in the presence of multiple objects. Nevertheless, the range resolution of FMCW radar is determined by the swept bandwidth B according to $\Delta R = \frac{C}{2B}$ [34], where C is the speed of light. Achieving high resolution therefore requires a very large sweep bandwidth, which increases hardware cost, design complexity, and regulatory challenges.

VII. CONCLUSIONS

This paper introduced a contactless vital-sign monitoring framework based on commercial off-the-shelf (COTS) 77 GHz mmWave FMCW radar. A comprehensive signal processing pipeline was designed to extract respiration- and heartbeat-induced micromotions from raw radar returns by leveraging range–Doppler–phase decomposition, phase unwrapping, band-pass filtering, and statistical estimation. The proposed approach was validated through a pilot study with a diverse cohort of participants, achieving mean absolute errors of 0.95 breaths/minute for respiration rate and 2.00 beats/minute for heart rate, without requiring subject-specific calibration. These findings demonstrate that commodity mmWave radar can deliver accurate, robust, and cross-subject physiological monitoring, offering a low-cost and unobtrusive alternative to conventional contact-based sensing modalities.

REFERENCES

- [1] R. A. Cox and C. Z. Torres, “Acute Heart Failure in Adults,” *Puerto Rico Health Sciences Journal*, vol. 23, no. 4, pp. 265–271, 2004.
- [2] D. Fan, A. Ren, N. Zhao, X. Yang, Z. Zhang, S. A. Shah, F. Hu, and Q. H. Abbasi, “Breathing Rhythm Analysis in Body Centric Networks,” *IEEE Access*, vol. 6, pp. 32 507–32 513, 2018.
- [3] A. R. Fekr, M. Janidarmian, K. Radecka, and Z. Zilic, “Respiration Disorders Classification With Informative Features for m-Health Applications,” *IEEE Journal of Biomedical and Health Informatics*, vol. 20, no. 3, pp. 733–747, May 2016.
- [4] Q. He, E. Yang, S. Fang, and S. Zhao, “Revisiting wireless breath and crowd inference attacks with defensive deception,” *IEEE/ACM Transactions on Networking*, vol. 32, no. 6, pp. 4976–4988, 2024.
- [5] —, “Honeybreath: An ambush tactic against wireless breath inference,” in *Mobile and Ubiquitous Systems: Computing, Networking and Services*, S. Longfei and P. Bodhi, Eds. Cham: Springer Nature Switzerland, 2023, pp. 203–226.
- [6] Philips Healthcare, “Intellivue x3 patient monitor,” Product page, 2025, accessed: 2025-08-31. [Online]. Available: <https://www.usa.philips.com/healthcare/product/HC867030/intellivue-x3-patient-monitor>
- [7] Masimo, “Mightysat rx fingertip pulse oximeter,” Masimo website, 2025, accessed: 2025-08-31. [Online]. Available: <https://www.masimo.com/products/wearables/mightysatrx/>
- [8] Garmin, “Sports & fitness activity and fitness trackers,” Product category page, 2025, accessed: 2025-08-31. [Online]. Available: <https://www.garmin.com/en-US/c/sports-fitness/activity-fitness-trackers/>
- [9] Wellue, “Ekg/e .ecg machines – portable, wireless & touchscreen options,” Product collection page, 2025, accessed: 2025-08-31. [Online]. Available: <https://getwellue.com/collections/ekg-machine>
- [10] M. Nakatsuka, H. Iwatani, and J. Katto, “A Study on Passive Crowd Density Estimation Using Wireless Sensors,” in *The 4th Intl. Conf. on Mobile Computing and Ubiquitous Networking (ICMU 2008)*. Citeseer, 2008.
- [11] Y. Yuan, J. Zhao, C. Qiu, and W. Xi, “Estimating Crowd Density in an RF-Based Dynamic Environment,” *IEEE Sensors Journal*, vol. 13, no. 10, pp. 3837–3845, 2013.
- [12] C. Xu, B. Firner, R. S. Moore, Y. Zhang, W. Trappe, R. Howard, F. Zhang, and N. An, “SCPL: Indoor Device-Free Multi-Subject Counting and Localization Using Radio Signal Strength,” in *Proceedings of the 12th International Conference on Information Processing in Sensor Networks*, ser. IPSN ’13. New York, NY, USA: Association for Computing Machinery, 2013, p. 79–90.
- [13] W. Xi, J. Zhao, X.-Y. Li, K. Zhao, S. Tang, X. Liu, and Z. Jiang, “Electronic Frog Eye: Counting Crowd Using WiFi,” in *IEEE INFOCOM 2014 - IEEE Conference on Computer Communications*, 2014, pp. 361–369.
- [14] A. Hanif, M. Iqbal, and F. Munir, “WiSpy: Through-Wall Movement Sensing and Person Counting Using Commodity WiFi Signals,” in *2018 IEEE SENSORS*, 2018, pp. 1–4.
- [15] X. Guo, B. Liu, C. Shi, H. Liu, Y. Chen, and M. C. Chuah, “WiFi-Enabled Smart Human Dynamics Monitoring,” in *Proceedings of the 15th ACM Conference on Embedded Network Sensor Systems*, ser. SenSys ’17. New York, NY, USA: Association for Computing Machinery, 2017.
- [16] S. Liu, Y. Zhao, and B. Chen, “WiCount: A Deep Learning Approach for Crowd Counting Using WiFi Signals,” in *2017 IEEE International Symposium on Parallel and Distributed Processing with Applications and 2017 IEEE International Conference on Ubiquitous Computing and Communications (ISPA/IUCC)*, 2017, pp. 967–974.
- [17] F. Adib, H. Mao, Z. Kabelac, D. Katabi, and R. C. Miller, “Smart Homes That Monitor Breathing and Heart Rate,” in *Proc. of ACM Conference on Human Factors in Computing Systems (CHI)*, 2015, p. 837–846.
- [18] L. Chen, J. Xiong, X. Chen, S. I. Lee, D. Zhang, T. Yan, and D. Fang, “LungTrack: Towards Contactless and Zero Dead-Zone Respiration Monitoring with Commodity RFIDs,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 3, no. 3, pp. 79:1–79:22, Sep 2019.
- [19] P. Hillyard, A. Luong, A. S. Abrar, N. Patwari, K. Sundar, R. Farney, J. Burch, C. Porucznik, and S. H. Pollard, “Experience: Cross-Technology Radio Respiratory Monitoring Performance Study,” in *Proc. of the 24th Annual International Conference on Mobile Computing and Networking*, ser. MobiCom ’18. New York, NY, USA: Association for Computing Machinery, 2018, p. 487–496.
- [20] W. Jia, H. Peng, N. Ruan, Z. Tang, and W. Zhao, “WiFind: Driver Fatigue Detection with Fine-Grained Wi-Fi Signal Features,” *IEEE Transactions on Big Data*, vol. 6, no. 2, pp. 269–282, 2020.
- [21] X. Liu, J. Cao, S. Tang, J. Wen, and P. Guo, “Contactless Respiration Monitoring Via Off-the-Shelf WiFi Devices,” *IEEE Transactions on Mobile Computing*, vol. 15, no. 10, pp. 2466–2479, Oct 2016.
- [22] J. Liu, Y. Wang, Y. Chen, J. Yang, X. Chen, and J. Cheng, “Tracking Vital Signs During Sleep Leveraging Off-the-shelf WiFi,” in *Proc. of ACM International Symp. on Mobile Ad Hoc Networking and Computing (MobiHoc)*, 2015, pp. 267–276.
- [23] N. Patwari, L. Brewer, Q. Tate, O. Kaltiokallio, and M. Bocca, “Breathfinding: A Wireless Network That Monitors and Locates Breathing in a Home,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 8, no. 1, pp. 30–42, Feb 2014.
- [24] Y. Zeng, D. Wu, R. Gao, T. Gu, and D. Zhang, “FullBreathe: Full Human Respiration Detection Exploiting Complementarity of CSI Phase and Amplitude of WiFi Signals,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 2, no. 3, Sep. 2018.
- [25] H. Wang, D. Zhang, J. Ma, Y. Wang, Y. Wang, D. Wu, T. Gu, and B. Xie, “Human Respiration Detection with Commodity WiFi Devices: Do User Location and Body Orientation Matter?” in *Proc. of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp)*, 2016, pp. 25–36.
- [26] X. Wang, K. Niu, J. Xiong, B. Qian, Z. Yao, T. Lou, and D. Zhang, “Placement Matters: Understanding the Effects of Device Placement for WiFi Sensing,” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 6, no. 1, 2022.
- [27] A. D. Droitcour, O. Boric-Lubecke, and G. T. A. Kovacs, “Signal-to-Noise Ratio in Doppler Radar System for Heart and Respiratory Rate Measurements,” *IEEE Transactions on Microwave Theory and Techniques*, vol. 57, no. 10, pp. 2498–2507, 2009.

- [28] M. Ascione, A. Buonanno, M. D'Urso, L. Angrisani, and R. Schiano Lo Moriello, "A New Measurement Method Based on Music Algorithm for Through-the-Wall Detection of Life Signs," *IEEE Transactions on Instrumentation and Measurement*, vol. 62, no. 1, pp. 13–26, 2013.
- [29] C. Li, V. M. Lubecke, O. Boric-Lubecke, and J. Lin, "A Review on Recent Advances in Doppler Radar Sensors for Noncontact Healthcare Monitoring," *IEEE Transactions on Microwave Theory and Techniques*, vol. 61, no. 5, pp. 2046–2060, 2013.
- [30] T. Rahman, A. T. Adams, R. V. Ravichandran, M. Zhang, S. N. Patel, J. A. Kientz, and T. Choudhury, "DoppleSleep: A Contactless Unobtrusive Sleep Sensing System Using Short-Range Doppler Radar," in *Proc. of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing*, ser. UbiComp '15. New York, NY, USA: Association for Computing Machinery, 2015, p. 39–50.
- [31] S. Venkatesh, C. R. Anderson, N. V. Rivera, and R. M. Buehrer, "Implementation and Analysis of Respiration-rate Estimation Using Impulse-based UWB," in *2005 IEEE Military Communications Conference (MILCOM)*, 2005, pp. 3314–3320 Vol. 5.
- [32] J. Salmi and A. F. Molisch, "Propagation Parameter Estimation, Modeling and Measurements for Ultrawideband MIMO Radar," *IEEE Transactions on Antennas and Propagation*, vol. 59, no. 11, pp. 4257–4267, 2011.
- [33] K. Van Loon, M. Breteler, L. Van Wolfwinkel, A. R. Leyssius, S. Kossen, C. Kalkman, B. van Zaane, and L. Peelen, "Wireless Non-invasive Continuous Respiratory Monitoring with FMCW Radar: A Clinical Validation Study," *Journal of clinical monitoring and computing*, vol. 30, no. 6, pp. 797–805, 2016.
- [34] F. Adib, Z. Kabelac, D. Katabi, and R. C. Miller, "3D Tracking via Body Radio Reflections," in *11th USENIX Symposium on Networked Systems Design and Implementation (NSDI 14)*. Seattle, WA: USENIX Association, Apr. 2014, pp. 317–329.
- [35] N. Molinaro, E. Schena, S. Silvestri, F. Bonotti, D. Aguzzi, E. Viola, F. Buccolini, and C. Massaroni, "Contactless vital signs monitoring from videos recorded with digital cameras: An overview," *Frontiers in Physiology*, vol. 13, p. 801709, 2022.
- [36] R. Sinhal, K. Singh, and A. Shankar, "Estimating vital signs through non-contact video-based approaches: A survey," in *2017 International Conference on Recent Innovations in Signal processing and Embedded Systems (RISE)*, 2017, pp. 139–141.
- [37] D. Qiao, A. H. Ayesha, F. Zulkernine, N. Jaffar, and R. Masroor, "Revise: Remote vital signs measurement using smartphone camera," *IEEE Access*, vol. 10, pp. 131 656–131 670, 2022.
- [38] P. M. Sanderson, B. Brecknell, S. Leong, S. Klueber, E. Wolf, A. Hickling, T.-L. Tang, E. Bell, S. Y. Li, and R. G. Loeb, "Monitoring vital signs with time-compressed speech," *Journal of Experimental Psychology: Applied*, vol. 25, no. 4, p. 647, 2019.
- [39] L. Wang, W. Li, K. Sun, F. Zhang, T. Gu, C. Xu, and D. Zhang, "Loear: Push the range limit of acoustic sensing for vital sign monitoring," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 6, no. 3, Sep. 2022.
- [40] O. Kaltiokallio, H. Yigitler, R. Jantti, and N. Patwari, "Non-invasive Respiration Rate Monitoring Using a Single COTS TX-RX Pair," in *Proc. of the 13th International Symp. on Information Processing in Sensor Networks (IPSN)*, 2014, pp. 59–69.
- [41] N. Patwari, J. Wilson, S. Ananthanarayanan, S. K. Kasera, and D. R. Westenskow, "Monitoring Breathing via Signal Strength in Wireless Networks," *IEEE Transactions on Mobile Computing*, vol. 13, no. 8, pp. 1774–1786, Aug 2014.
- [42] H. Abdelnasser, K. A. Harras, and M. Youssef, "UbiBreathe: A Ubiquitous Non-Invasive WiFi-Based Breathing Estimator," in *Proc. of ACM International Symp. on Mobile Ad Hoc Networking and Computing (MobiHoc)*, 2015, p. 277–286.
- [43] X. Liu, J. Cao, S. Tang, and J. Wen, "Wi-Sleep: Contactless Sleep Monitoring via WiFi Signals," in *2014 IEEE Real-Time Systems Symposium*, Dec 2014, pp. 346–355.
- [44] X. Wang, C. Yang, and S. Mao, "PhaseBeat: Exploiting CSI Phase Data for Vital Sign Monitoring with Commodity WiFi Devices," in *2017 IEEE 37th International Conference on Distributed Computing Systems (ICDCS)*, June 2017, pp. 1230–1239.
- [45] X. Wang, C. Yang, and S. Mao, "TensorBeat: Tensor Decomposition for Monitoring Multiperson Breathing Beats with Commodity WiFi," *ACM Trans. Intell. Syst. Technol.*, vol. 9, no. 1, Sep. 2017.
- [46] F. Zhang, D. Zhang, J. Xiong, H. Wang, K. Niu, B. Jin, and Y. Wang, "From Fresnel Diffraction Model to Fine-Grained Human Respiration Sensing with Commodity Wi-Fi Devices," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 2, no. 1, Mar. 2018.
- [47] L. Gui, C. Ma, B. Sheng, Z. Guo, J. Cai, and F. Xiao, "In-Home Monitoring Sleep Turnover Activities and Breath Rate via WiFi Signals," *IEEE Systems Journal*, 2022.
- [48] J. Liu, Y. Zeng, T. Gu, L. Wang, and D. Zhang, "WiPhone: Smartphone-based Respiration Monitoring Using Ambient Reflected WiFi Signals," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 5, no. 1, mar 2021.
- [49] Y. Yin, X. Yang, J. Xiong, S. I. Lee, P. Chen, and Q. Niu, "Ubiquitous Smartphone-Based Respiration Sensing With Wi-Fi Signal," *IEEE Internet of Things Journal*, vol. 9, no. 2, pp. 1479–1490, 2022.
- [50] Z. Yang, P. H. Pathak, Y. Zeng, X. Liran, and P. Mohapatra, "Monitoring vital signs using millimeter wave," ser. MobiHoc '16. New York, NY, USA: Association for Computing Machinery, 2016, p. 211–220.
- [51] Y. Wu, H. Ni, C. Mao, J. Han, and W. Xu, "Non-intrusive human vital sign detection using mmwave sensing technologies: A review," *ACM Trans. Sen. Netw.*, vol. 20, no. 1, Nov. 2023.
- [52] Z. Yang, P. H. Pathak, Y. Zeng, X. Liran, and P. Mohapatra, "Vital sign and sleep monitoring using millimeter wave," *ACM Trans. Sen. Netw.*, vol. 13, no. 2, Apr. 2017.
- [53] F. Wang, F. Zhang, C. Wu, B. Wang, and K. J. R. Liu, "Vimo: Multiperson vital sign monitoring using commodity millimeter-wave radio," *IEEE Internet of Things Journal*, vol. 8, no. 3, pp. 1294–1307, 2021.
- [54] B. Zhang, B. Jiang, R. Zheng, X. Zhang, J. Li, and Q. Xu, "Pi-vimo: Physiology-inspired robust vital sign monitoring using mmwave radars," *ACM Trans. Internet Things*, vol. 4, no. 2, May 2023.
- [55] A. Hyvärinen and E. Oja, "Independent component analysis: algorithms and applications," *Neural Netw.*, vol. 13, no. 4–5, p. 411–430, May 2000.
- [56] Texas Instruments, "Iwr1843boost evaluation module (boosterpack) for 77 ghz mmwave radar," <https://www.ti.com/tool/IWR1843BOOST>, 2025, accessed: 2025-08-31.
- [57] —, "Dca1000evm real-time data-capture adapter evaluation module," Product page, 2025, accessed: 2025-08-31. [Online]. Available: <https://www.ti.com/tool/DCA1000EVM>
- [58] Federal Communications Commission and others, "Revision of Part 15 of the Commission's Rules Regarding Ultra-wideband Transmission Systems," *First Report and Order, FCC 02-48*, 2002.
- [59] L. Lampe and K. Witrals, "Challenges and Recent Advances in IR-UWB System Design," in *Proc. of 2010 IEEE International Symposium on Circuits and Systems*, 2010, pp. 3288–3291.
- [60] O. Boric-Lubecke, V. M. Lubecke, A. D. Droitcour, B.-K. Park, and A. Singh, *Doppler Radar Physiological Sensing*. Wiley Online Library, 2016.
- [61] Yanming Xiao, J. Lin, O. Boric-Lubecke, and V. M. Lubecke, "Frequency-tuning Technique for Remote Detection of Heartbeat and Respiration Using Low-power Double-sideband Transmission in the ka-band," *IEEE Transactions on Microwave Theory and Techniques*, vol. 54, no. 5, pp. 2023–2032, 2006.